

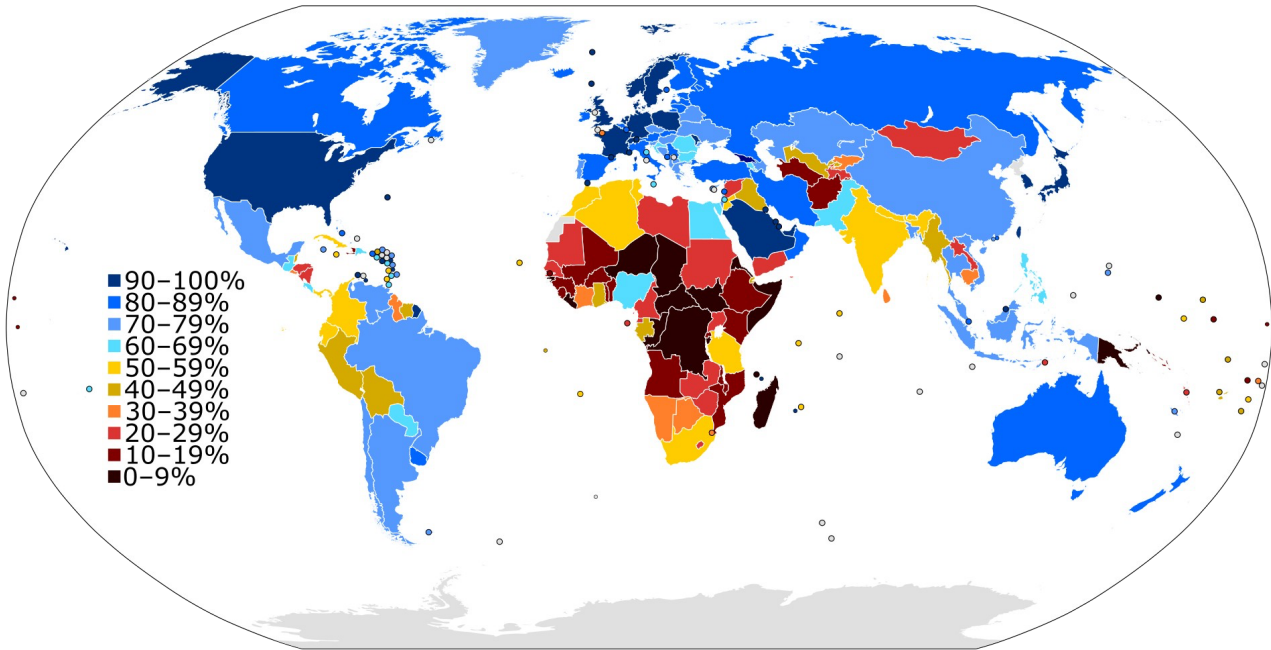


Politechnika Krakowska
im. Tadeusza Kościuszki

Dynamic routing - Classification, RIP

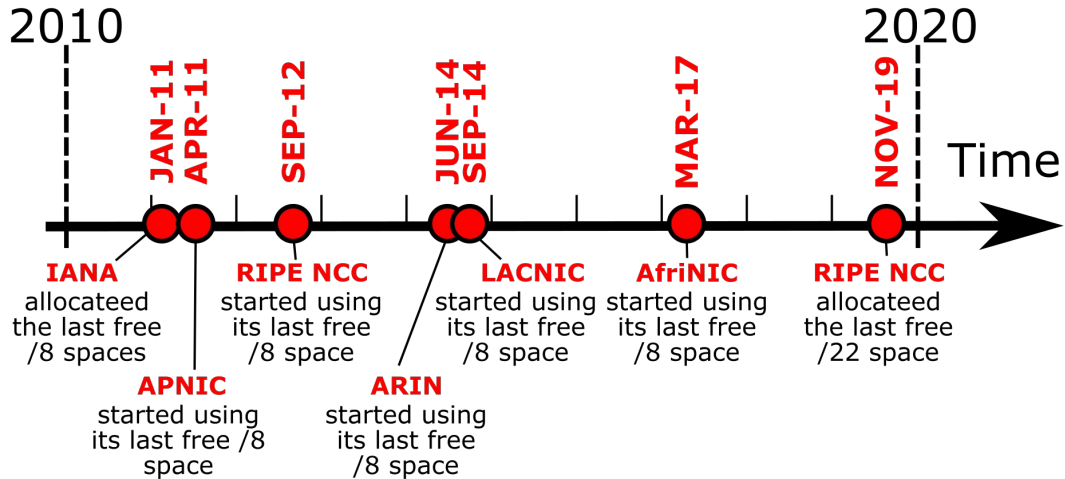
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Politechnika Krakowska

Development of the Internet



https://en.wikipedia.org/wiki/List_of_countries_by_number_of_Internet_users

IPv4 exhaustion



https://en.wikipedia.org/wiki/IPv4_address_exhaustion

Classification of dynamic routing protocols

Routing types

Static Routing

- Knowledge of network topology required
- Manual route entries by the system administrator

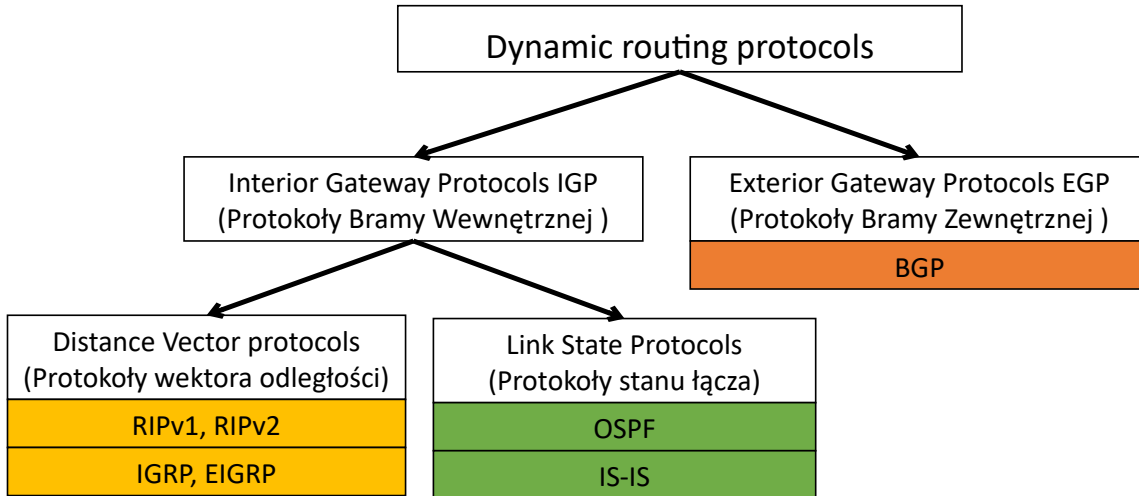
Dynamic routing

- No knowledge of network topology necessary
- Routes determine router because of dynamic routing protocols

Static vs dynamic routing

Feature	Static routing	Dynamic routing
Configuration complexity	It grows as the network grows	Independent of the size of the network
Topology changes	They require an administrator	Automatic
Security	+	-
Resource usage	+	-
Predictability	Always the same	Depends on configuration

Dynamic routing protocols



Konwergencja protokołów routingu dynamicznego

Konwergencję osiąga się, gdy zawartość wszystkich routerów jest spójna.

Czas zbieżności to czas potrzebny routerom na udostępnienie informacji, obliczenie najlepszej ścieżki i aktualizację tablic routingu.

Routery działają niezależnie od siebie.
Mówi się, że routery dążą do osiągnięcia konsensusu.

Classification of dynamic routing protocols

	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
	Distance vector routing protocols (protokoły r. wektora odległości)		Link-state routing protocols (protokoły r. stanu-łącza)		Path-vector r. p. (p.wektora-trasy)
Class (klasowe)	RIP	IGRP			EGP
Classless (bezklasowe)	RIPv2	EIGRP	OSPFv2	IS-IS	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGPv4 for IPv6

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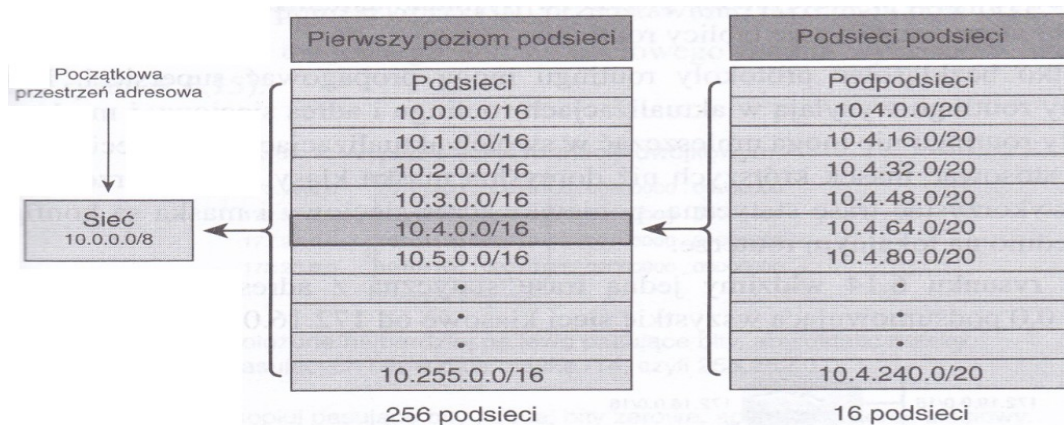
Class addressing

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	<u>0</u> .0.0.0 - <u>127</u> .255.255.255	255.0.0.0	127	16.777.214
B	1 0 _ _	<u>128</u> .0.0.0 - <u>191</u> .255.255.255	255.255.0.0	16.382	65.534
C	1 1 0 _	<u>192</u> .0.0.0 - <u>223</u> .255.255.255	255.255.255.0	2.097.150	254
D	1 1 1 0	<u>224</u> .0.0.0 - <u>239</u> .255.255.255	for multicast transmission		
E	1 1 1 1	<u>240</u> .0.0.0 - <u>255</u> .255.255.255	Reserved for IETF		



	(protokoły r. wektora odległości)	(protokoły r. stanu-łacza)	(p. wektora-trasy)		
Class (klasowe)	RIP	IGRP	EGP		
Classless (bezklasowe)	RIPv2	EIGRP	OSPFv2	IS-IS	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGPv4 for IPv6

Classless addressing (CIDR, VLSM)



	Distance vector routing protocols (protokoły r. wektora odległości)		Link-state routing protocols (protokoły r. stanu-łączy)		External g. p. (p.b.zewnętrznej)
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Classification of dynamic routing protocols

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Autonomous system

The Internet is based on the concept of autonomous systems.

An autonomous system (routing domain) is a collection of routers that remain under a common administration. It has a consistent routing scheme.

An example of autonomous systems:

- Internet Service Providers (ISP),
- CENIC in California,
- CYFRONET in Kraków,

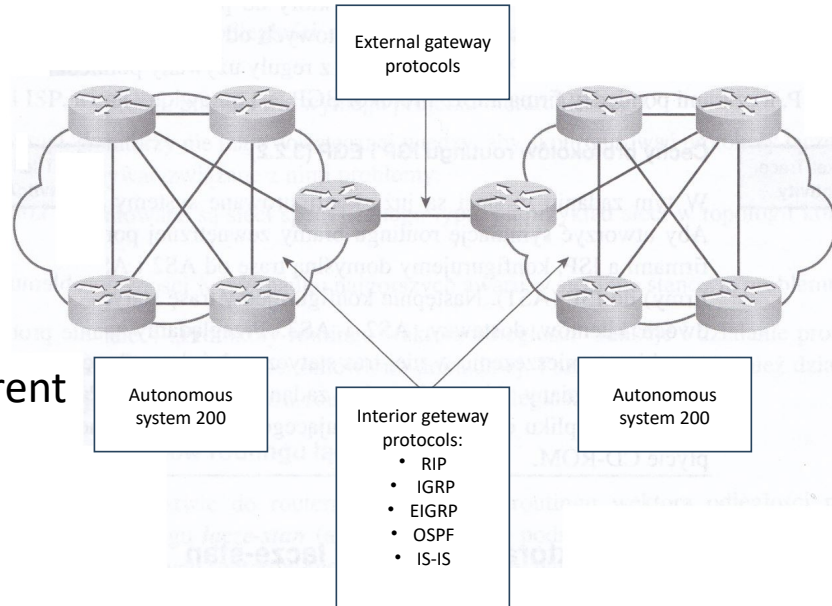
IGP and EGP gateway protocols

Interior gateway protocols

- Used for routing within an autonomous system,

Exterior gateway protocols

- Used for routing between different autonomous systems,



Classification of dynamic routing protocols

	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (protokoły zewnętrznej)
	Distance vector routing protocols (protokoły r. wektora odległości)		Link-state routing protocols (protokoły r. stanu-łączy)		Path-vector r. p. (protokół r. wektora-trasy)
Class (klasowe)	RIP	IGRP			EGP
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Distance vector routing

In distance vector routing, the router knows only:

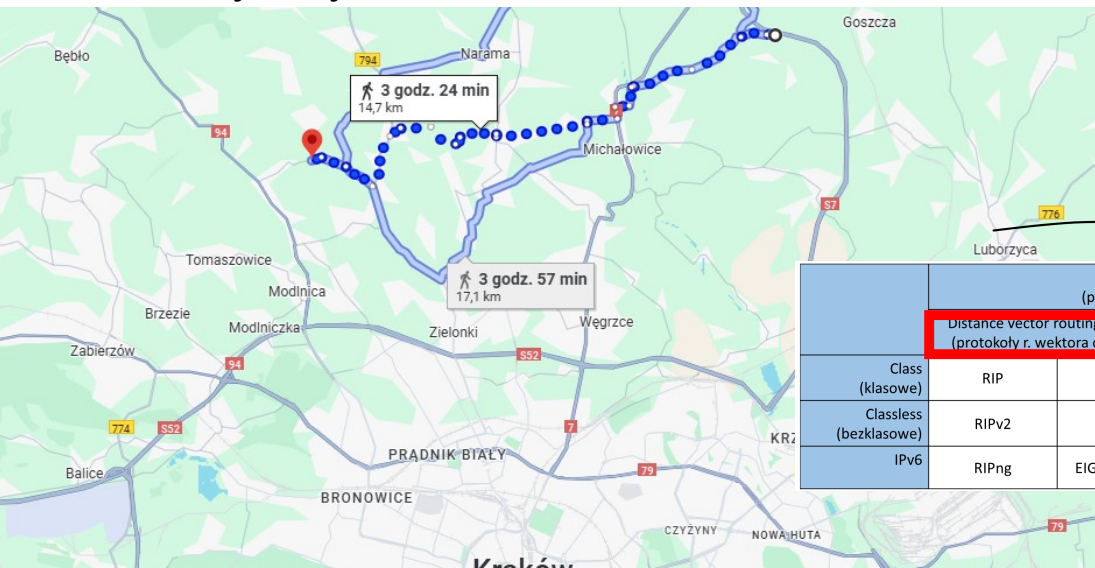
- Direction to send packets (interfaces),
- Distance to target network - number of hops,

The distance is defined by a **metric**.

	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
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Routing wektora odległości

Jak najbliżej



	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
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Link-state routing

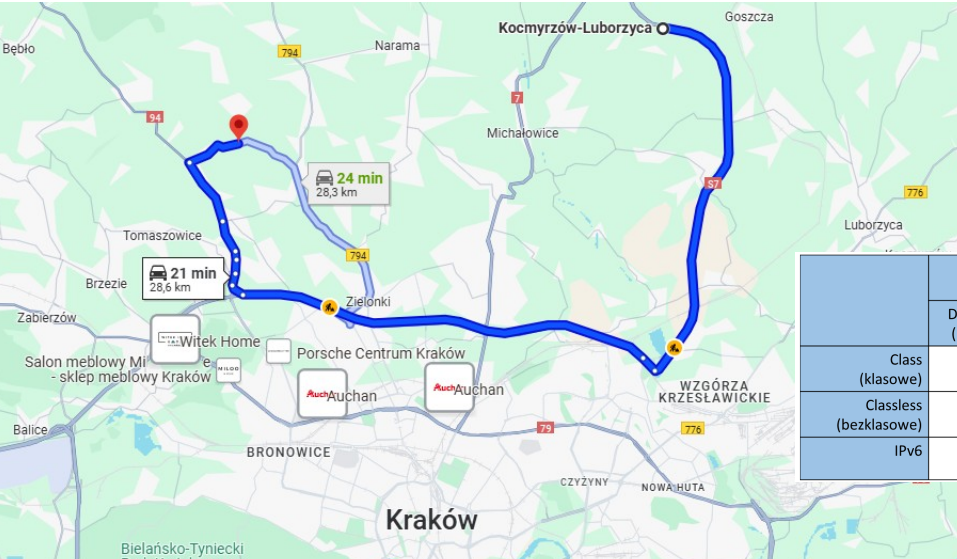
Link-state routing

- It is based on Edsger Dijkstra's shortest path first (SPF) algorithm,
- The shortest path does not have to be the path with the fewest jumps

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Routing łącze-stan

Jak najkorzystniej



	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
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Path vector routing

Routing route vector

- Used in external BGP gateway protocols
- The route vector routing uses various attributes for measurement.

At the level of Internet Service Provider (ISP), choosing the fastest route is often more important than the number of hops.

	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
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Overview of dynamic routing protocols on the example of RIP

Routing protocols

	Gateway protocols internal				External gateway protocols
	Distance vector routing protocols		Link-state Routing protocols		Path-vector R. Protocols
Class	RIP	IGRP			EGP
Classless	RIPv2	EIGRP	OSPFv2	IS-IS	BGPv4
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Protokoły RIP

RIPv1 (RFC 1058 z roku 1988)

- Praktycznie wyszedł z użycia,

RIPv2 (RFC 1723 z roku 1994)

- Unowocześniona wersja protokołu RIPv1

RIPng (RFC 2080 z roku 1997)

- Modyfikacja protokołów RIP pod IPv6

	Interior gateway protocols (protokoły bramy wewnętrznej)				External g. p. (p.b.zewnętrznej)
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Messages in dynamic routing

To update their routing tables, the routers send **MESSAGES** among themselves, the task of which is to inform each other about the state of the network.

The content of the message varies depending on the routing protocol.

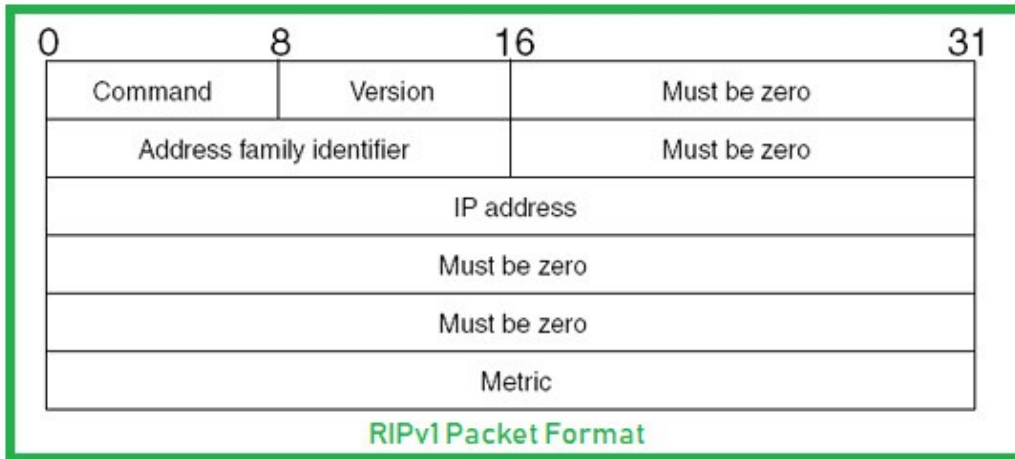
RIP - message encapsulation

Data link frame header	IP packet header	UDP segment header	RIP message (512B; up to 25 routes)
Data link frame Source MAC address Destination MAC address			
	IP packet Source IP address Destination IP address • RIPv1 – broadcast 255.255.255.255 • RIPv2 – multicast 224.0.0.9		
		UDP segment Source port = 520, Destination port = 520	
			RIP message

RIPv1

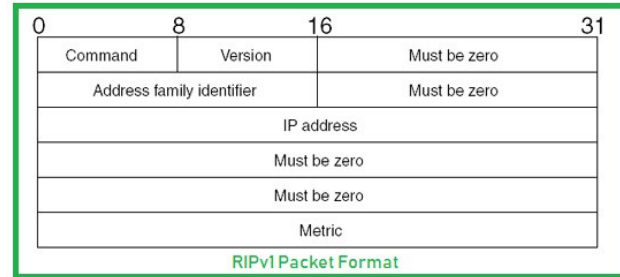
RIPv1 – message content

RIPv1 frame construction

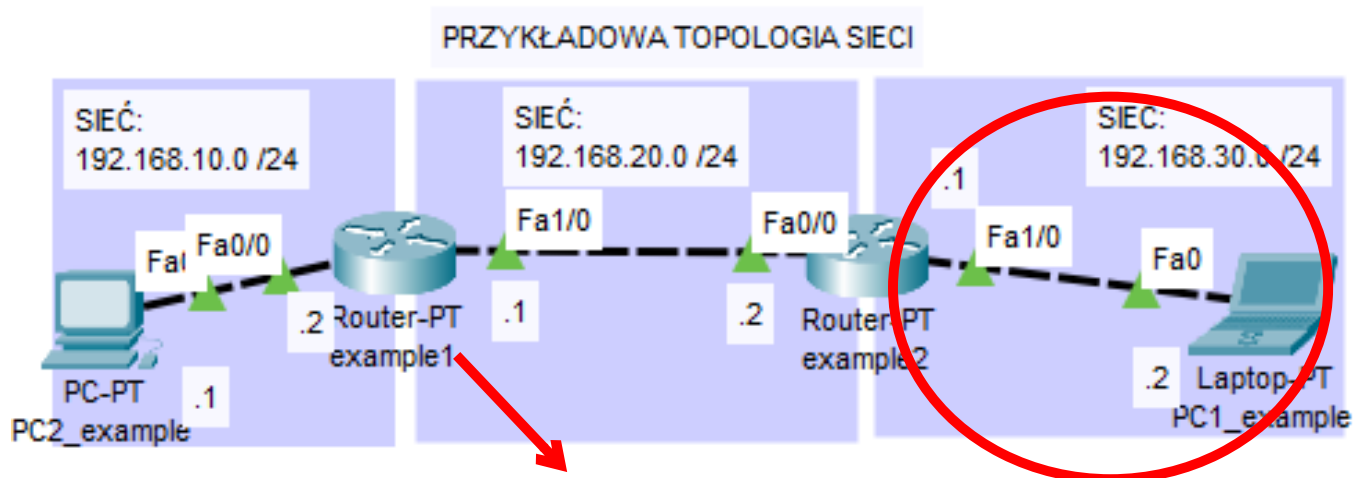


RIPv1 – message content

- 1) Why 0s in the RIPv1 message?
- 2) Where's the netmask?
- 3) How often are these shipped messages?



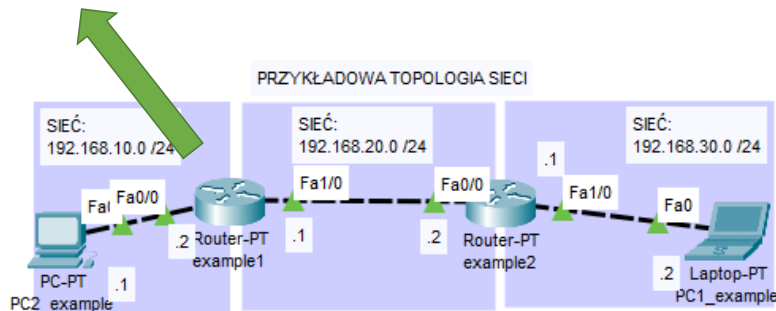
Sample topology – routing



THIS ROUTER DOESN'T KNOW THE NETWORK 192.168.30.0/24

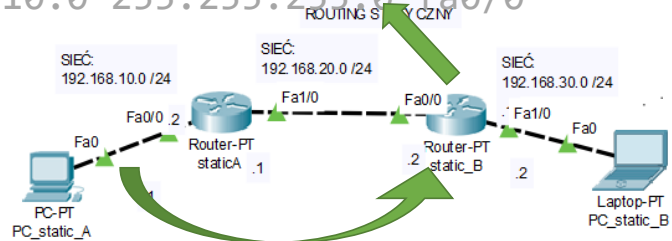
Przykładowa topologia - konfiguracja adresów

```
Router> enable
Router# configure terminal
Router(config)# hostname RXY
RXY(config)# interface fa0/0 ! shorter
RXY(config-if)# ip address 192.168.20.2 255.255.255.0
RXY(config-if)# no shutdown
RXY(config-if)# end
RXY# show running-config
RXY# show ip interface fa0/0
RXY# disable
RXY> exit
```



Static routing – configuration in IOS

```
R_B> enable
R_B# show interfaces
R_B# show ip interface brief
R_B# show ip route
R_B# configure terminal
R_B(config)# ip route 192.168.10.0 255.255.255.0 192.168.20.1
R_B(config)# ! no ip route 192.168.10.0 255.255.255.0 fa0/0
R_B(config)# end
R_B# show ip interface brief
R_B# show ip route
R_B# disable
```



Static Routing - wyświetlanie trasy

S 192.168.10.0/24 [1/0] via 192.168.20.1

1

2

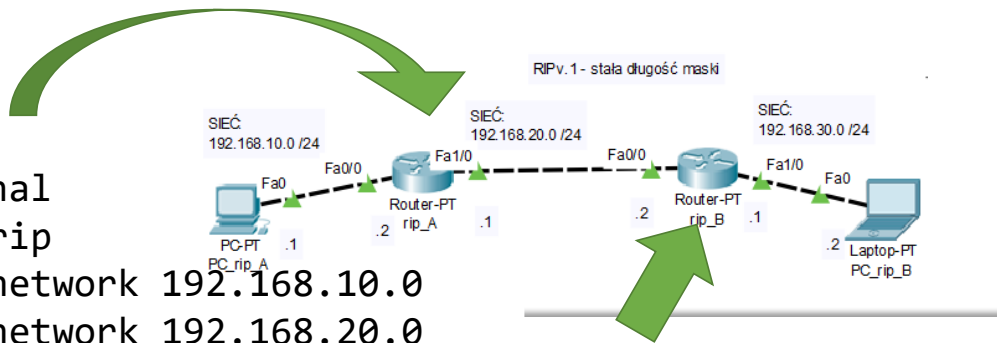
3

4

1. **S** - kod tablicy routingu dla trasy statycznej
2. **192.168.10.0/24** - adres sieciowy trasy docelowej
3. **[1/0]** – metryka / odległość administracyjna
4. **via 192.168.20.1** - adres IP routera następnego przeskoku, w tym przypadku adres IP interfejsu FastEthernet1/0 drugiego routera

RIPv1 – configuration in IOS

```
R_A> enable
R_A# show ip route
R_A# debug ip rip
R_A# configure terminal
R_A(config)# router rip
R_A(config-router)# network 192.168.10.0
R_A(config-router)# network 192.168.20.0
R_A(config-router)# exit
R_A# show interfaces
R_A# show ip interface brief
R_A# show ip route
R_A# exit
```



```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

R 192.168.10.0/24 [120/1] via 192.168.20.1, 00:00:25, FastEthernet0/0
C 192.168.20.0/24 is directly connected, FastEthernet0/0
C 192.168.30.0/24 is directly connected, FastEthernet1/0

Router>|
```

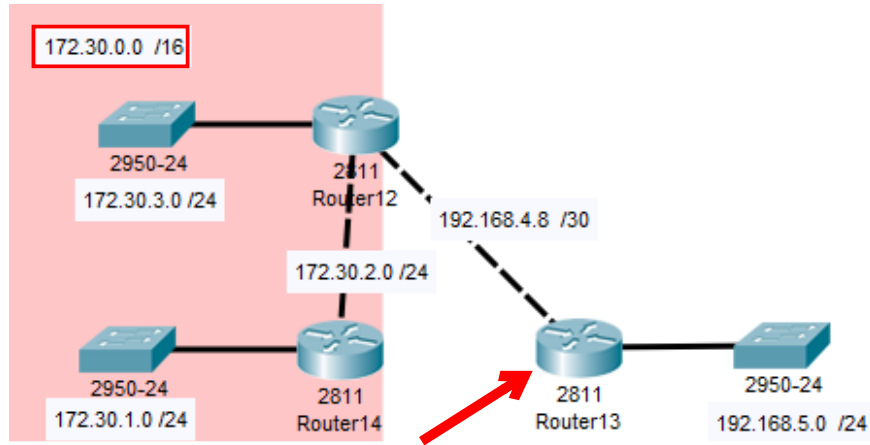
RIPv1 – track record

```
R    192.168.10.0/24 [120/1] via 192.168.20.1,
      00:00:25, FastEthernet 0/0
```

1. R – routing table code for RIP
2. 192.168.10.0/24 – remote network address,
3. [120/1] – administrative distance / metric (number of hops)
4. via 192.168.20.1 – next-hop router's IP address,
5. 00:00:25 – counter,
6. FastEthernet 0/0 – local interface that leads to the remote network.

RIP protocols -
discontinuous networks, autosumization

Auto summary of networks



Automatic summarization takes place here to network 172.30.0.0/16

Advantages and disadvantages of automatic summarization

Advantages of automatic summarization:

- Smaller routing updates are sent,
- Using a single route gives a faster routing table lookup process,

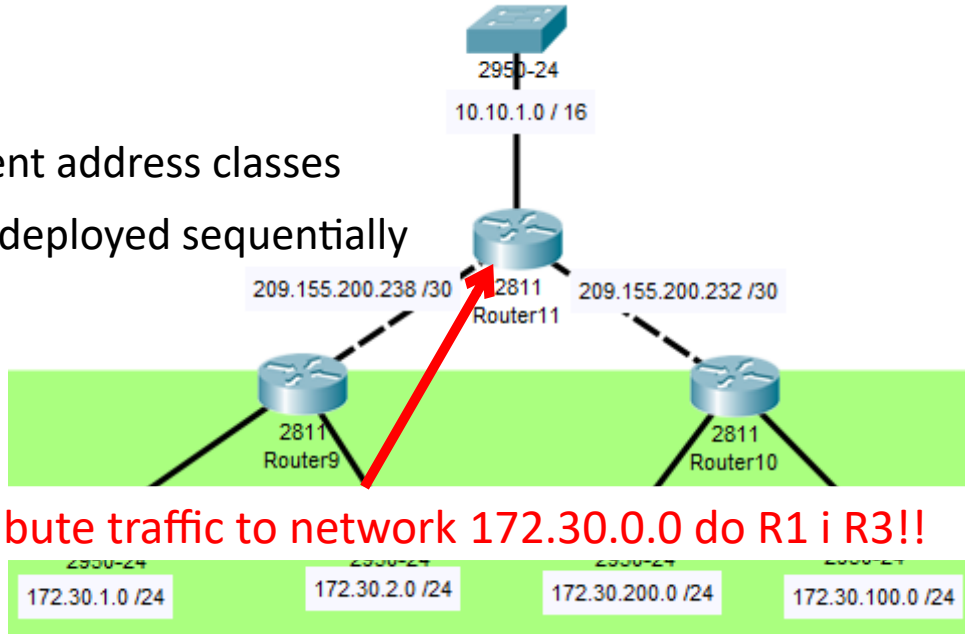
Disadvantages of automatic summarization:

- RIPv1 in non-contiguous topologies does not converge

Discontinuous network

It takes place online when

- Subnets belong to different address classes
- The networks cannot be deployed sequentially



In RIPv1, router R2 will distribute traffic to network 172.30.0.0 do R1 i R3!!

RIPv2

RIPv2 – założenia

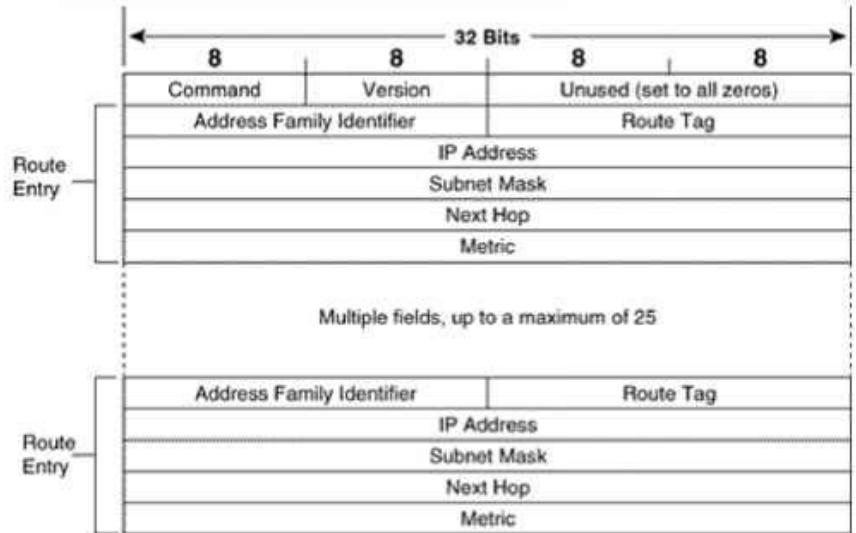
RIPv2 implementuje technologie takie jak: VLSM i CIDR:

- VLSM (variable length subnet masks, bezklasowy protokół routingu dynamicznego)
 - Zasady podziału sieci na podsieci o zmiennej długości maski,
 - Bardziej efektywne wykorzystanie przestrzeni adresowej IPv4,
- CIDR (Classless Inter-Domain Routing, bezklasowy routing między domenami):
 - Agregacja tras w celu zmniejszenia zmniejszenie tablicy routingu,
 - Redukcja ilości informacji o trasach przenoszonych przez routery rdzeniowe.

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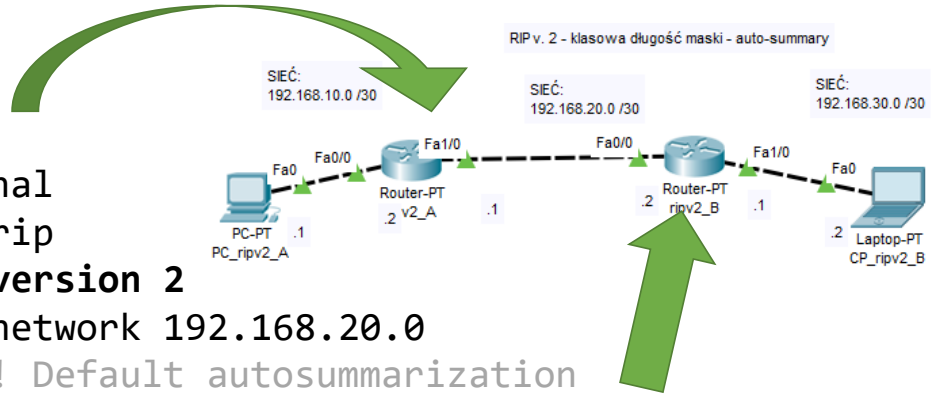
RIPv2 – message content

RIPv2 frame format



RIPv2 – configuration with auto-summary

```
RXY> enable
RXY# show ip route
RXT# debug ip rip
RXT# configure terminal
RXY(config)# router rip
RXY(config-router)# version 2
RXY(config-router)# network 192.168.20.0
RXY(config-router)# ! Default autosummarization
RXY(config-router)# exit
RXY# show interfaces
RXY# show ip interface brief
RXY# show ip route
RXY# exit
```

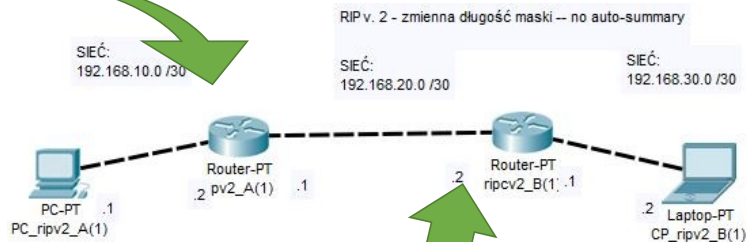


Gateway of last resort is not set

```
R 192.168.10.0/24 [120/1] via 192.168.20.1, 00:00:20, F
192.168.20.0/30 is subnetted, 1 subnets
C 192.168.20.0 is directly connected, FastEthernet0/
192.168.30.0/30 is subnetted, 1 subnets
C 192.168.30.0 is directly connected, FastEthernet1/
```

RIPv2 – configuration without auto-summary

```
RXY> enable
RXY# show ip route
RXT# debug ip rip
RXT# configure terminal
RXY(config)# router rip
RXY(config-router)# version 2
RXY(config-router)# network 192.168.20.0
RXY(config-router)# no auto-summary
RXY(config-router)# exit
RXY# show interfaces
RXY# show ip interface brief
RXY# show ip route
RXY# exit
```



Gateway of last resort is not set

```
R    192.168.10.0/30 is subnetted, 1 subnets
C    192.168.10.0 [120/1] via 192.168.20.1, 00:00:14, FastEthernet0/0
C    192.168.20.0/30 is subnetted, 1 subnets
C    192.168.20.0 is directly connected, FastEthernet0/0
C    192.168.30.0/30 is subnetted, 1 subnets
C    192.168.30.0 is directly connected, FastEthernet1/0
```

RIPv1 vs RIPv2 - IOS comparison

```
Router>
Router>sh ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 2 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
Interface Send Recv Triggered RIP Key Chain
FastEthernet0/0 1 2 1
FastEthernet1/0 1 2 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.20.0
  192.168.30.0
Passive Interface(s):
Routing Information Sources:
  Gateway Distance Last Update
  192.168.30.1 120 00:00:00
Distance: (default is 120)
Router>
```

```
Router#
Router#sh ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 16 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
Interface Send Recv Triggered RIP Key Chain
FastEthernet0/0 2 2
FastEthernet1/0 2 2
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.20.0
  192.168.30.0
Passive Interface(s):
Routing Information Sources:
  Gateway Distance Last Update
  192.168.30.1 120 00:00:00
Distance: (default is 120)
Router#
```

RIP protocols - timers

Network updates - triggering messages

In distance vector protocols (RIPv1, RIPv2, IGRP), routers regularly send routing tables to their neighbors.

In RIP:

- every 30 seconds
- after initiation



Rimers in RIP protocols

```
Router#sh ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
192.168.10.0/30 is subnetted, 1 subnets  
R    192.168.10.0 [120/1] via 192.168.20.1, 00:00:11, FastEthernet0/0  
192.168.20.0/30 is subnetted, 1 subnets  
C    192.168.20.0 is directly connected, FastEthernet0/0  
192.168.30.0/30 is subnetted, 1 subnets  
C    192.168.30.0 is directly connected, FastEthernet1/0
```

```
Router#
```

Sources

1. Akademia sieci CISCO Exploration;
Semestr 1 Podstawy Sieci, Mark A. Dyne;
2. Akademia sieci CISCO Exploration;
Semestr 2 Protokoły i koncepcje routingu, Rick Graziani;
3. Akademia sieci CISCO Exploration; Semestr
3 Przełączanie sieci LAN i sieci bezprzewodowe, Wayne Lewis;
4. CCNA 200-125 Zostań administratorem sieci komputerowych,
Adam Józefiok;

